



Shenzhen Hi-Link Electronic Co., Ltd.

HLK-LD2410S Operation Guide

Content

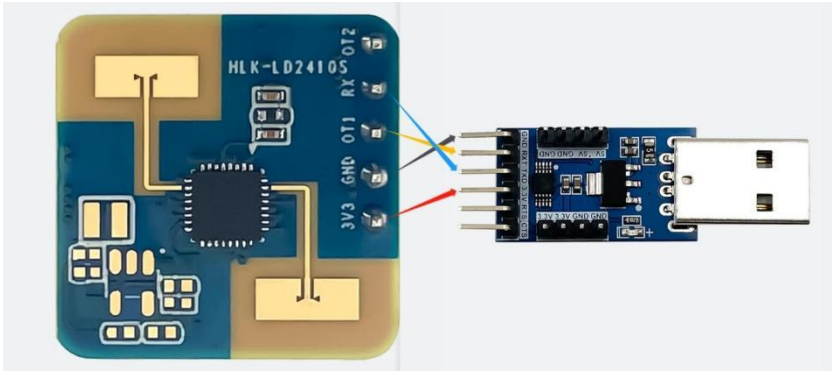
I. Wiring Tutorial	1
II. Test whether the module is working properly	1
2.1. Install driver file	1
2.2. View serial port data	2
2.2.1. Default reported data	2
2.2.2. Check if sending and receiving commands are normal	3
2.3. Simple Test	3
2.3.1. Default parameter	3
2.3.2. Simple Test	4
III. Notes and Common Issues (.....	5
3.1. Firmware and host computer correspondence	5
3.2. Detailed Explanation of Older Host Computer Parameters	5
3.2.1. Minimum detection distance and maximum detection distance	5
3.2.2. Why can the module's detection range be set to 8.4 meters when it is set to 8 meters?	6
3.2.3. Why is the minimum distance gate set to 1.4 meters, yet the target can still be detected within 1.4 meters?	6
3.2.4. What is a distance gate?	6
3.2.5. What is the corresponding relationship between distance and threshold?	7
3.2.6. Trigger threshold and trigger signal-to-noise ratio threshold? Hold threshold and hold signal-to-noise ratio threshold? What are they?	7
3.2.7. What are the status reporting frequency and distance reporting frequency?	7

3.3. Detailed Explanation of New Host Computer Parameters	7
3.3.1. What are the trigger threshold and hold threshold?	8
3.3.2. What is meant by reading/writing sensor settings?	8
3.3.3. What is the purpose of generating automatic thresholds?	8
3.3.4. What is the time period when no one reports?	9
3.4. How to upgrade firmware	9
3.5. Frequently Asked Questions	10
3.5.1. Serial port device failed to open?	10
3.5.2. Failed to write/read parameters ?	11
3.5.3. How to switch to standard data mode and parse standard data ?	11
3.5.4. How to parse standard data?	12
IV. Debugging Parameter	13
4.1. Use automatically generated threshold adjustment parameters	13
4.1.1. Use the host computer to automatically generate the threshold	13
4.1.2. Automatically generate threshold	13
4.2. Fine-tuning parameter	14

1. Wiring tutorial

Pin -> Corresponding wiring	
2410S pin	CH340 (Serial Port Small Board)
3V3	3.3V
GND	GND
OT1 (defined as TX)	RX
RX	TX

OT2 is defined as the OUT output pin and can be left unconnected.



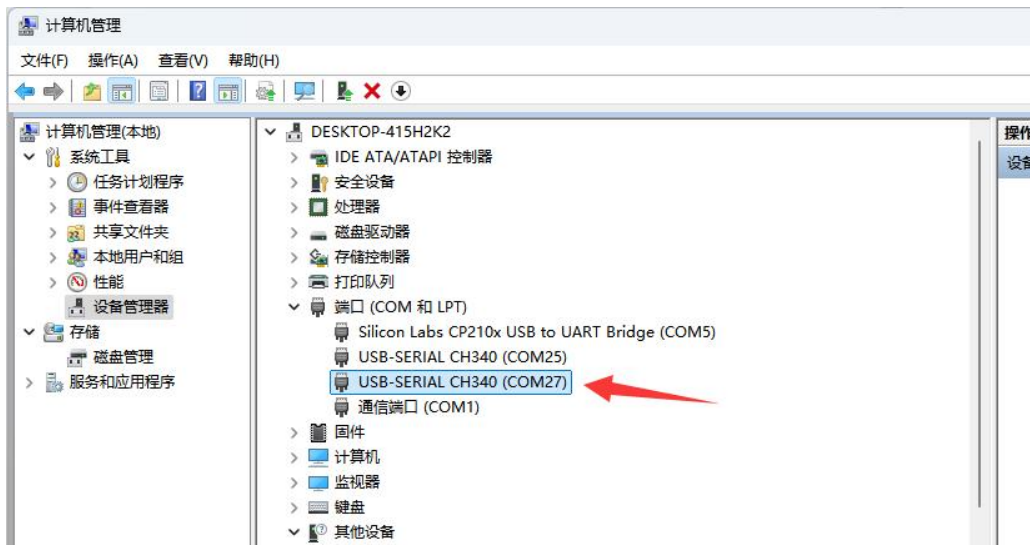
2. Test whether the module is working properly

2.1 Install driver files

(1) Open the driver file, click install, and the driver installation will be completed after installation.



(2) After the driver is installed, connect the cable and you can see the device management port as shown in the figure below.



2.2 View serial port data

2..2.1 Default reported data

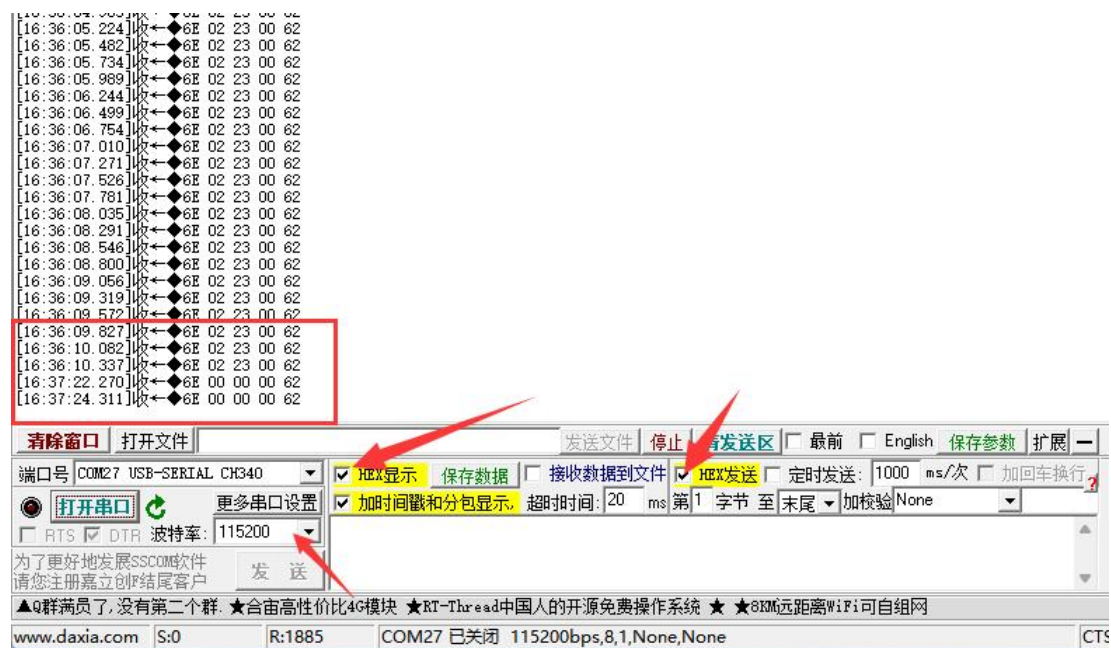
The 2410S serial port has a default baud rate of 115200, 1 stop bit, and no parity bit. The default reported data is shown in the figure below. The data is in little-endian format, with a distance of 0023, which is 35 in decimal.

[16:36:10.082] Received ←◆6E 02 23 00 62

Frame header: 6E Target status: 02 (Someone is there) Target distance: 23 00 (35cm) Frame tail: 62

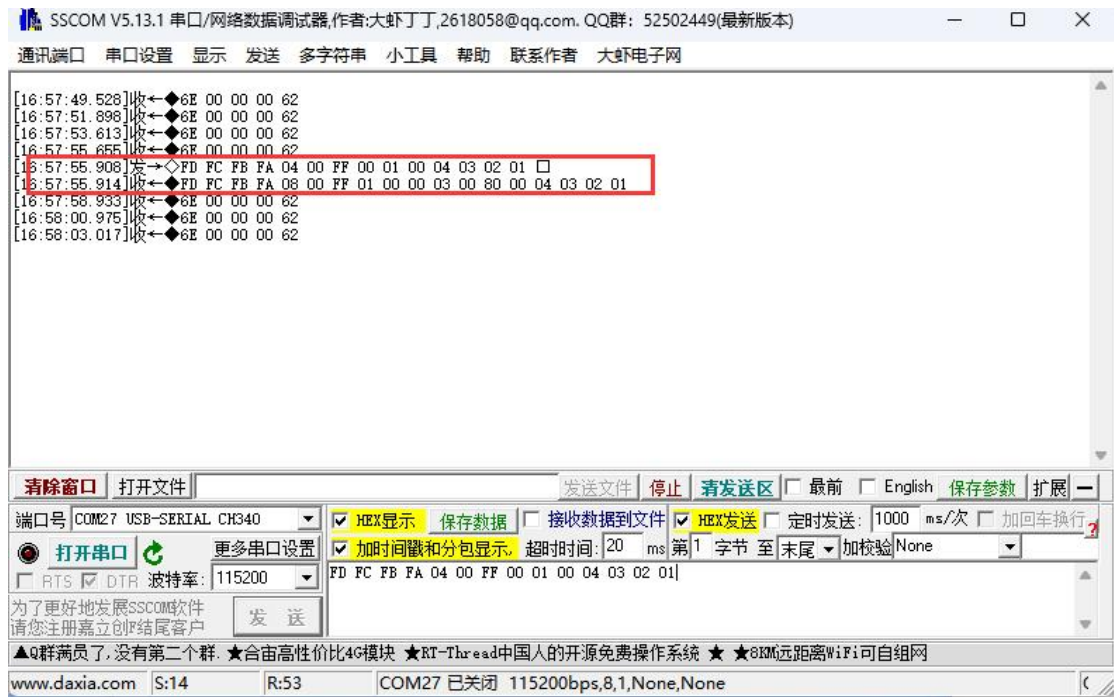
[16:37:22.270] Receive ←◆6E 00 00 00 62

Frame header: 6E Target status: 00 (No one) Target distance: 23 00 (35cm) Frame tail: 62



2.2.2 Check if sending and receiving commands are normal.

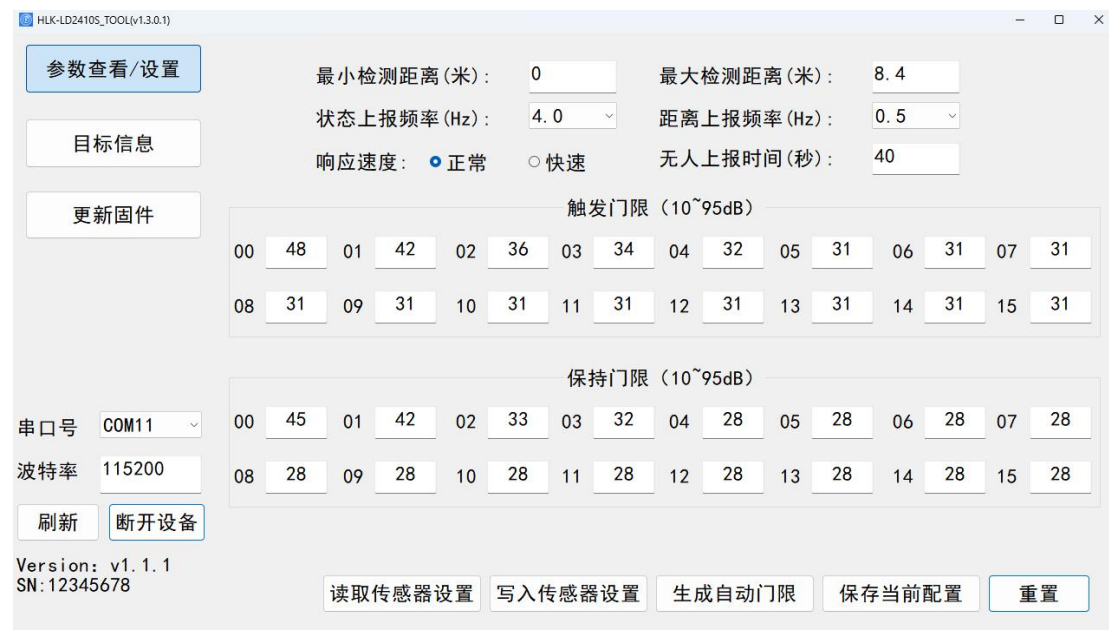
Send the enable configuration command (FD FC FB FA 04 00 FF 00 01 00 04 03 02 01) and check for a reply. Normally, data will stop being reported after the enable configuration command is replied to. You need to send parameter commands within 3 seconds. If no command is sent within 3 seconds, it will automatically exit the configuration mode.



Normal sending and receiving indicates that the module is working properly.

2.3 Simple test

2.3.1 Default parameters

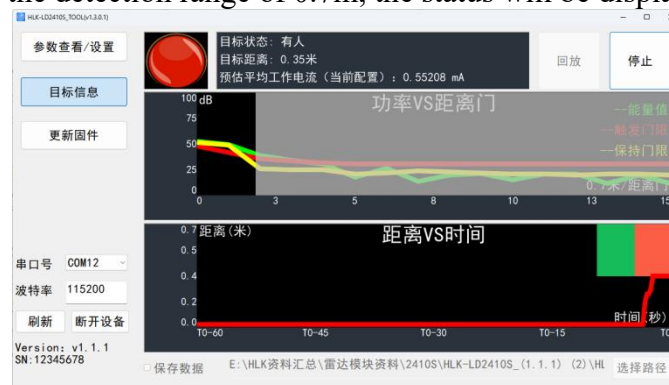


2.3.2 Simple test

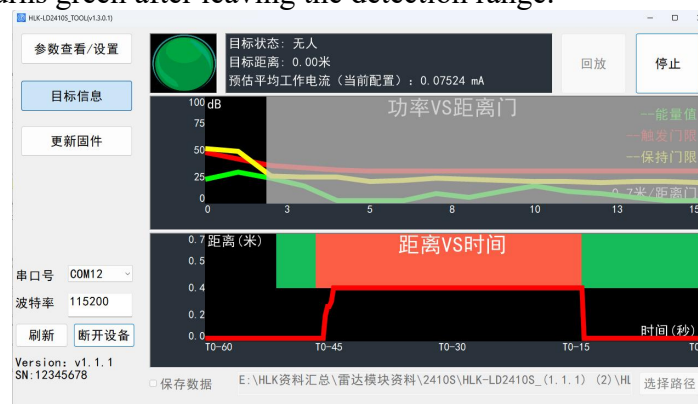
To test the radar's functionality, set the maximum detection range to 0.7 meters, the target disappearance delay to 10 seconds, the frequency to 8 Hz, and the response speed to fast. After modifying the parameters, write them back to the sensor settings. To check if the writing was successful, you can read the settings again.



At this point, enter the target information page and check the status changes. If a person is within the detection range of 0.7m, the status will be displayed in red.



The status turns green after leaving the detection range.



If the status changes, it means that the module is working normally.

3. Notes and Frequently Asked Questions

3.1 Firmware and host computer compatibility

The 2410S comes in two firmware versions: old firmware (V1.1.0 and V1.0.1) and new firmware (V1.1.1), corresponding to the old host computer (V1.2.0.15) and the new host computer (V1.3.0.1), respectively. If the corresponding host computer is not used, reading or writing parameters will fail. The corresponding relationships are shown in the table below:

2410S firmware	2410S host computer
Versions 1.1.0 and 1.0.1	V1.2.0.15
V1.1.1	V1.3.0.1

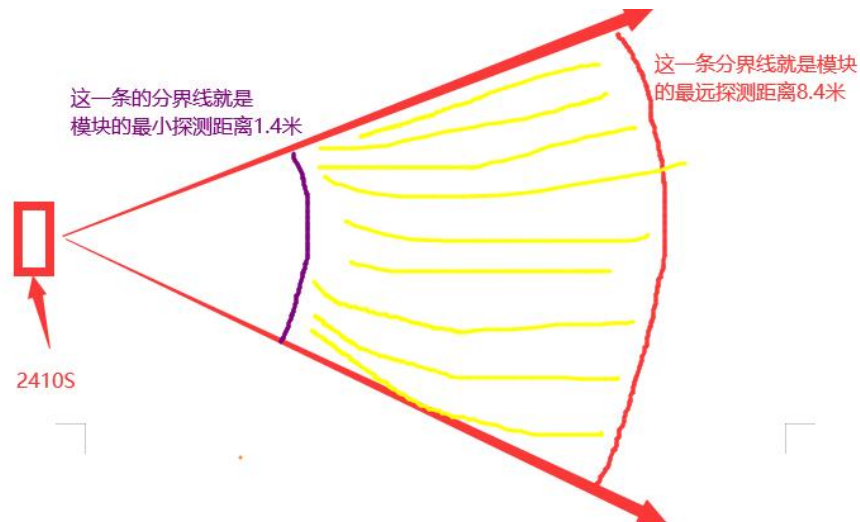
3.2 Detailed Explanation of Old Version Host Computer Parameters

The screenshot shows the HLK-LD2410S_TOOL(v1.2.0.15) software interface. It features a sidebar with buttons for '参数查看/设置' (Parameter View/Setting), '目标信息' (Target Information), and '更新固件' (Update Firmware). The main area contains several parameter settings:

- 最小检测距离(米)** (Minimum detection distance (m)): 0
- 最大检测距离(米)** (Maximum detection distance (m)): 2.80
- 状态上报频率** (Status report frequency): 8.0
- 距离上报频率** (Distance report frequency): 8.0
- 响应速度** (Response speed): ☐ 正常 (Normal), ☒ 快速 (Fast)
- 无人上报时间(秒)** (No one report time (s)): 10
- 触发门限 (10~95dB)** (Trigger threshold (10~95dB)): A row of 16 buttons with values: 00 48 01 42 02 36 03 34 04 32 05 31 06 31 07 31
- 触发信噪比门限 (5~63dB)** (Trigger SNR threshold (5~63dB)): A row of 16 buttons with values: 08 11 09 11 10 11 11 11 12 11 13 11 14 11 15 11
- 保持门限 (10~95dB)** (Hold threshold (10~95dB)): A row of 16 buttons with values: 00 45 01 42 02 36 03 33 04 32 05 28 06 28 07 28
- 保持信噪比门限 (5~63dB)** (Hold SNR threshold (5~63dB)): A row of 16 buttons with values: 08 10 09 10 10 10 11 10 12 10 13 10 14 10 15 10
- 串口号** (Serial port number): COM1
- 波特率** (Baud rate): 115200
- 刷新** (Refresh) and **连接设备** (Connect device) buttons.
- 信噪比: 目标需要比噪声高的倍** (SNR: The target needs to be higher than the noise by a factor of...)
- 读取传感器设置** (Read sensor settings), **写入传感器设置** (Write sensor settings), and **重置** (Reset) buttons.

3.2.1 Minimum detection distance and maximum detection distance

If the minimum detection distance is set to 1.4 and the maximum detection distance is set to 8.4, then the yellow shaded area is the detection area of the module.



3.2.2 Why can the module's detection range be set to 8.4 meters when it is 8 meters?

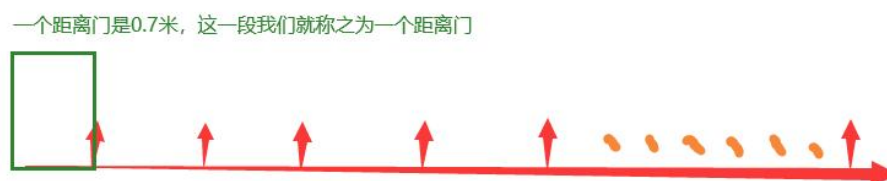
The distance parameter can only be set in multiples of 0.7. That is to say, if you need to detect a distance of 6 meters, then you need to set the maximum detection distance to 6.3. Of course, if you set it directly to 6 meters, after writing the parameter, it will be automatically saved as the closest number to your setting. That is to say, if you input 6, after writing the parameter, it will automatically become 6.3.

3.2.3 Why is the minimum distance threshold set to 1.4 meters? Can a distance of 1.4 meters still be detected?

It is generally not recommended to set a minimum distance gate; the default value should be kept at 0 because the module's detection distance may have errors and cannot completely block the detection distance of the preceding components.

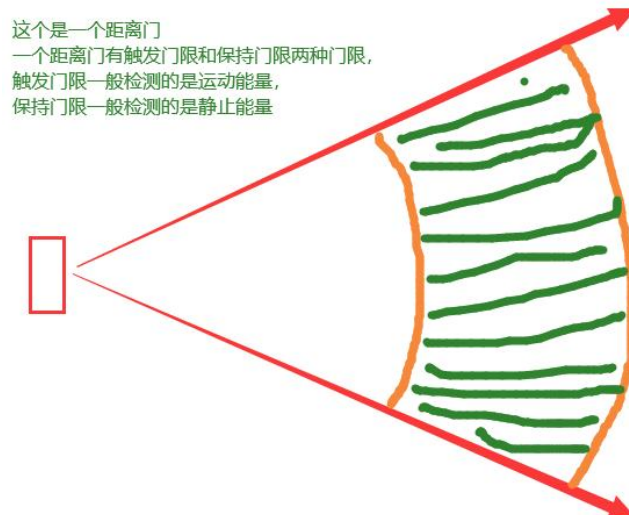
3.2.4 What is a distance gate?

Simply put, it's a distance with a total of 15 distance gates. Each distance gate is 0.7 meters, for a total distance of 10.5 meters. However, the 2410S module actually only has a detection distance of 8 meters, meaning that only the first 12 distance gates are effective.



3.2.5 What is the relationship between distance and threshold?

The resolution of a distance gate is 0.7 meters. This means that each distance gate has a corresponding threshold parameter. For example, there are 8 distance gates from 00 to 07. 00 corresponds to 0 meters, 01 corresponds to 0-0.7 meters, 02 corresponds to 0.7-1.4 meters, and so on. 07 corresponds to 4.2-4.9 meters.



3.2.6 Trigger threshold and trigger signal-to-noise ratio threshold? Hold threshold and hold signal-to-noise ratio threshold? What are they?

The trigger threshold and the trigger signal-to-noise ratio threshold are the same; both are threshold values. Initially, there were two interpretations because the parameter setting ranges for distance gates 00-07 and 08-15 were different. To distinguish them, these two interpretations have now been unified as the trigger threshold for easier understanding. The hold-hold threshold and the hold-signal-to-noise ratio threshold are similar.

3.2.7 What are the status reporting frequency and distance reporting frequency?

Status reporting frequency: The frequency at which the sensor reports whether the object is occupied or unoccupied .

Distance reporting frequency: The frequency at which the sensor reports the distance to the target when there are people present .

0.5Hz means uploading once every 2 seconds, and 8Hz means uploading 8 times every 1 second.

3.3 New version of host computer parameter details

HLK-LD2410S_TOOL(v1.3.0.1)

参数查看/设置

目标信息

更新固件

串口号: COM1

波特率: 115200

刷新

连接设备

最小检测距离(米): 0

最大检测距离(米): 5.6

状态上报频率(Hz): 4.0

距离上报频率(Hz): 4.0

响应速度: ☐ 正常 ☒ 快速

无人上报时间(秒): 40

触发门限 (10~95dB)

00	48	01	42	02	36	03	34	04	32	05	31	06	31	07	31
08	31	09	31	10	31	11	31	12	31	13	31	14	31	15	31

保持门限 (10~95dB)

00	45	01	42	02	36	03	33	04	32	05	28	06	28	07	28
08	28	09	28	10	28	11	28	12	28	13	28	14	28	15	28

读取传感器设置

写入传感器设置

生成自动门限

保存当前配置

重置

3.3.1 What are trigger thresholds and hold thresholds?

The new version of the host computer's **trigger threshold** corresponds to the **trigger threshold and trigger signal-to-noise ratio threshold of the old version**. Similarly, the **hold threshold** corresponds to the **hold threshold and hold signal-to-noise ratio threshold of the old version**. Simply put, a threshold is a set limit; when the energy value exceeds this threshold, it triggers. Radar works similarly: if the detected object's energy value exceeds the set threshold, it triggers; otherwise, it doesn't.

3.3.2 What are the read/write sensor settings?

Writing to the sensor means writing parameters, and reading from the sensor means reading parameters. After modifying the parameters, you can write the settings, and the set parameters will be written in. When you need to read the parameters, you use the "Read Sensor Settings" function.

3.3.3 What is the purpose of generating automatic thresholds?

This is used to eliminate environmental interference. It is generally used to detect background noise in unmanned environments. It is usually applicable under default parameters. If a person is not in the detection range but someone is still detected, this function can be used.

To give an analogy, if the trigger threshold for 01 is 42 and the hold threshold for 01 is 42, but the environment is empty, it will still keep triggering "someone is here".

After using this function to detect the background noise, the trigger threshold for 01 becomes 43 and the hold threshold for 01 becomes 43. When you detect it again, the state can change from "someone is here" to "no one is here", and it will not keep triggering "someone is here".

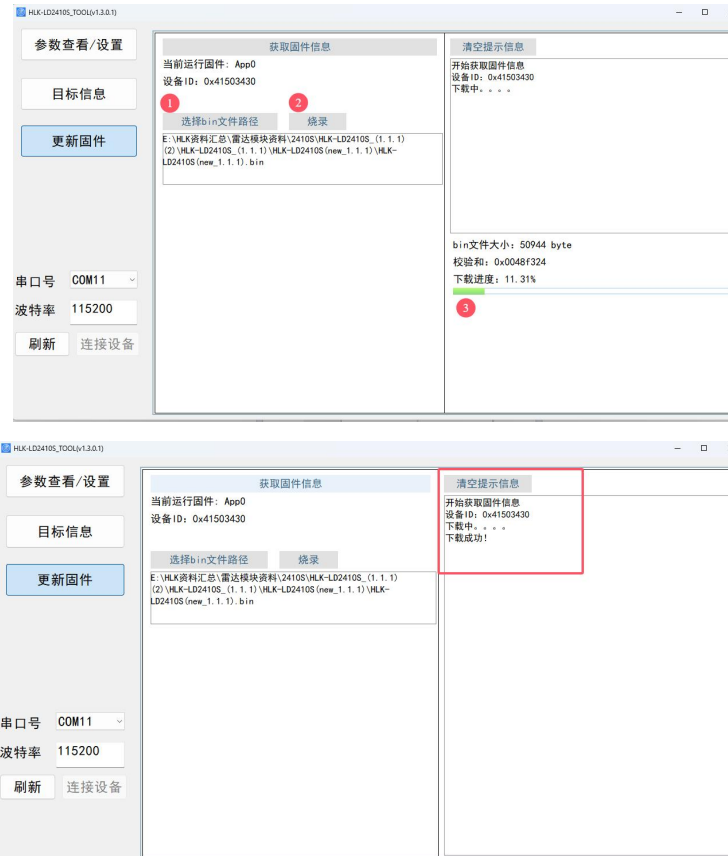
3.3.4 What is the time period when no one reports it?

Each distance gate has two thresholds: a trigger threshold and a hold threshold. Only when the energy values of all distance gates fall below all thresholds, and this time limit is exceeded, will the system transition from a occupied state to an unoccupied state. The default minimum is 10 seconds, which is currently limited by the internal algorithm and cannot be further reduced.

3.4 How to upgrade firmware via OTA

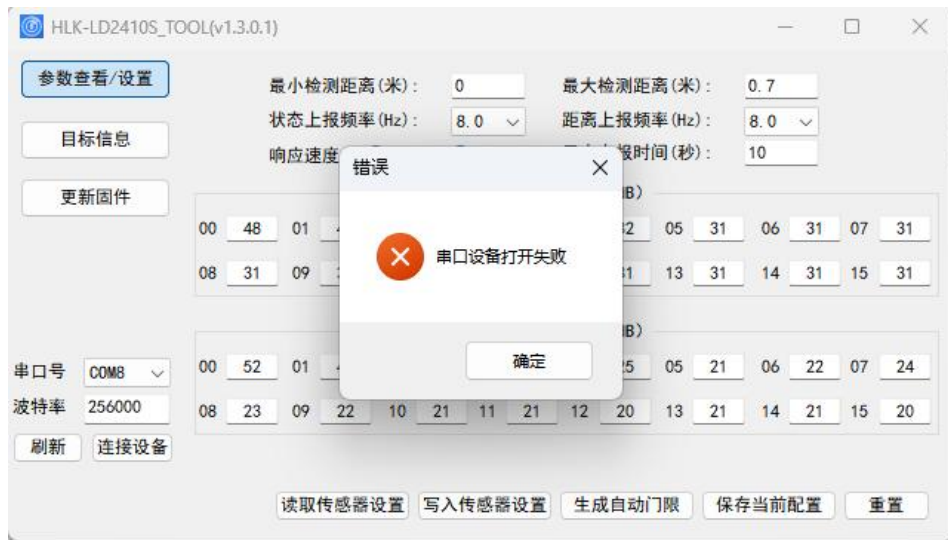
Select the corresponding serial port and baud rate. Note that you should not click "Connect Device" at this time. Next, click "Get Firmware Information", then select the path to the bin file, and finally click "Burn".





3.5 Frequently Asked Questions

3.5.1 Serial port device failed to open ?



Solution:

- (1) Check the serial port number and baud rate to ensure they are selected correctly. The default baud rate is 115200.
- (2) Check if the wiring is incorrect, and whether TX and RX are reversed.
- (3) Third-party serial port tools and host computers cannot be used simultaneously. When opening the host computer, the third-party serial port tool must be closed.
- (4) The module wiring may be loose. You can try unplugging and replugging the

serial port board and then restarting the host computer.

3.5.2 Failed to write/read parameters ?



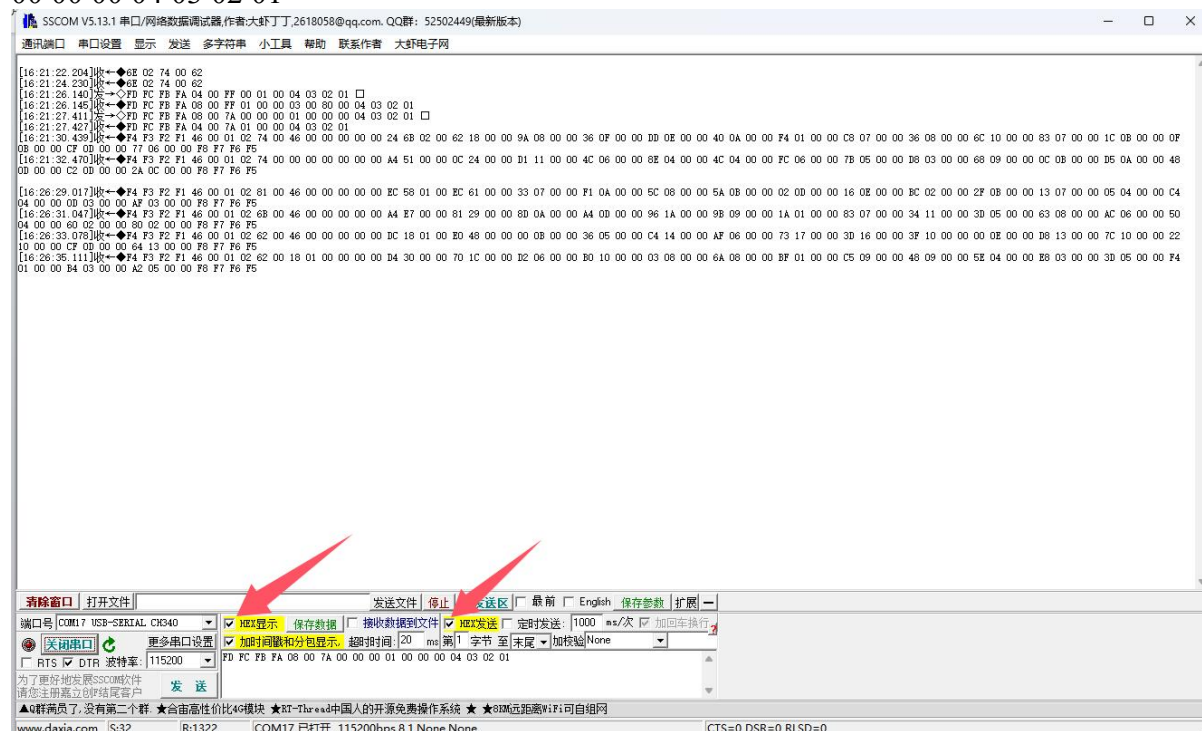
Solution:

Incorrect use of the host computer tool. Many users confuse the host computer tools for the 2410S and the 2420. In this case, you need to first confirm which module you have. Generally, there will be silkscreen markings on the module to identify it.

3.5.3 How to switch to standard data mode and parse standard data ?

(1) Send enable configuration command: FD FC FB FA 04 00 FF 00 01 00 04 03 02 01

(2) Send the standard data switching command: FD FC FB FA 08 00 7A 00 00 00 01 00 00 00 04 03 02 01



3.5.4 How to parse standard data?

门限参数说明： 假设 N 为上位机配置的参数，M 为串口配置的参数，对于上位机和串口设置

的参数转换关系是 $N = (10 * \log_{10} M)$ $M = 10^{\frac{N}{10}}$, 例如串口配置距离门 0 门限值为 65536, 对应上位机为 $(10 * \log_{10} 65536) \approx 48.16$ 。例如上位机设置的参数为 70, 对应串口配置参数为 $10^{\frac{70}{10}} \approx 10000000$, 指令转换 16 进制, 小端在前就为: **0x80969800**

[10:40:07.176]Receive ← ◆F4 F3 F2 F1 46 00 01 02 69 00 46 00 BC 1B 00 00
66 3B 00 00 0E 0D 00 00 F4 01 00 00 9C 01 00 00 64 00 00 00 64 00 00 00 DF 00 00
00 3C 01 00 00 5C 03 00 00 9C 01 00 00 D8 02 00 00 90 01 00 00 3C 01 00 00 58 02
00 00 84 03 00 00 F8 F7 F6 F5

Frame header : F4 F3 F2 F1

Intra-frame data length: 46 00

Data type: 01

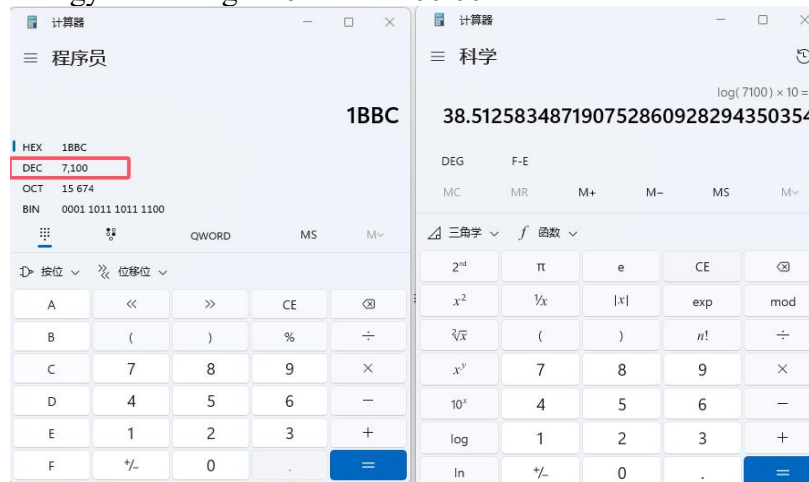
Target status : 02

Target distance: 69 00

Reserved bits: 46 00

Energy values of each distance gate

Energy value of gate 0: BC 1B 00 00. Calculation method is as follows:



Gate 1 Energy Value: 66 3B 00 00 Gate 2 Energy Value: 0E 0D 00 00

Energy value of Gate 3: F4 01 00 00 Energy value of Gate 4: 9C 01 00 00

Energy value of Gate 5: 64 00 00 00 Energy value of Gate 6: 64 00 00 00

Energy value of Gate 7: DF 00 00 00 Energy value of Gate 8: 3C 01 00 00

Energy value of door #9: 5C 03 00 00 Energy value of door #10: 9C 01 00 00

Energy value of Gate 11: D8 02 00 00 Energy value of Gate 12: 90 01 00 00

Energy value of Gate 13: 3C 01 00 00 Energy value of Gate 14: 58 02 00 00

Energy value of Gate 15: 84 03 00 00

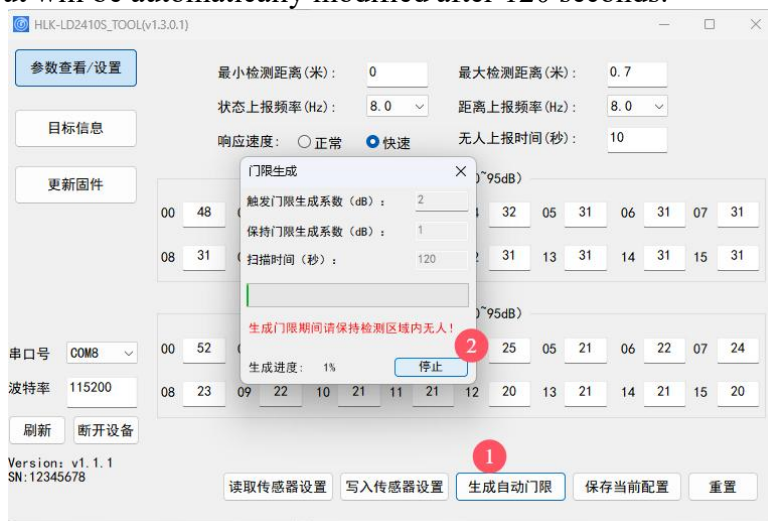
Frame end: F8 F7 F6 F5

4. Debug parameters

4.1 Use automatically generated threshold adjustment parameters

4.1.1 Use the host computer to automatically generate thresholds.

Keep the area within the detection range unattended, and begin generating an automatic threshold. The parameters for generating the threshold will not be modified by default, but will be automatically modified after 120 seconds.



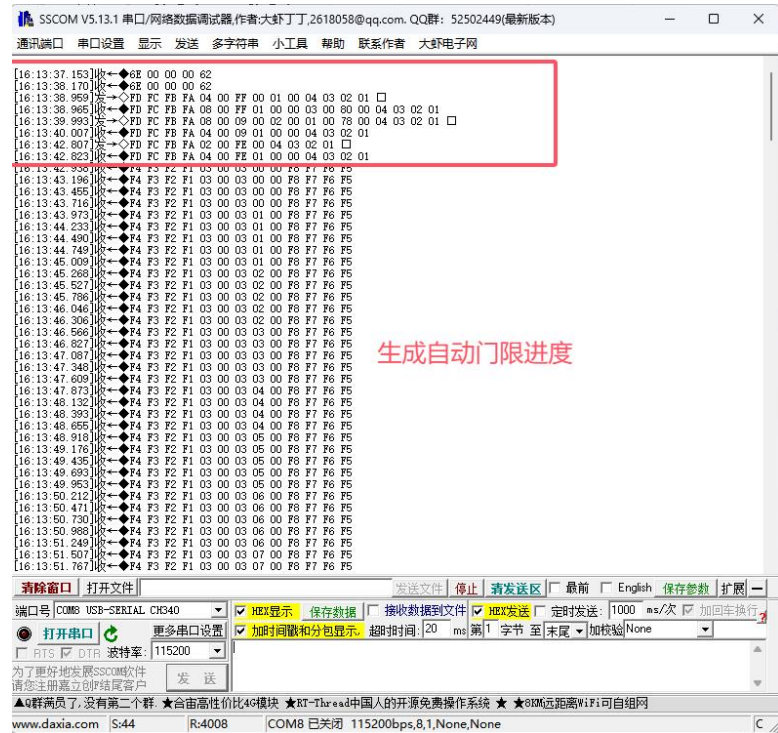
4.1.2 Automatically generate thresholds using serial port commands

Send the following three commands in sequence. Note that after sending the enable configuration, you need to send the next command within 3 seconds, otherwise it will be invalid.

FD FC FB FA 04 00 FF 00 01 00 04 03 02 01 (Enable configuration instruction)

FD FC FB FA 08 00 09 00 02 00 01 00 78 00 04 03 02 01 (Generate automatic threshold instruction)

FD FC FB FA 02 00 FE 00 04 03 02 01 (Disable enable configuration instructions)



4.2 Fine-tuning parameters

The fine-tuning parameters here are based on the automatically generated thresholds. According to the above understanding of the trigger threshold and hold threshold, the values of the trigger threshold and hold threshold are adjusted accordingly.

- (1) Increase the trigger threshold value if accidental activation is likely.
- (2) If it is difficult to trigger, lower the trigger threshold value.
- (3) If the situation cannot be maintained, the time for no one to report can be appropriately extended and the value of the maintenance threshold can be lowered.
- (4) When there are always people present, the value of the holding threshold can be appropriately increased.